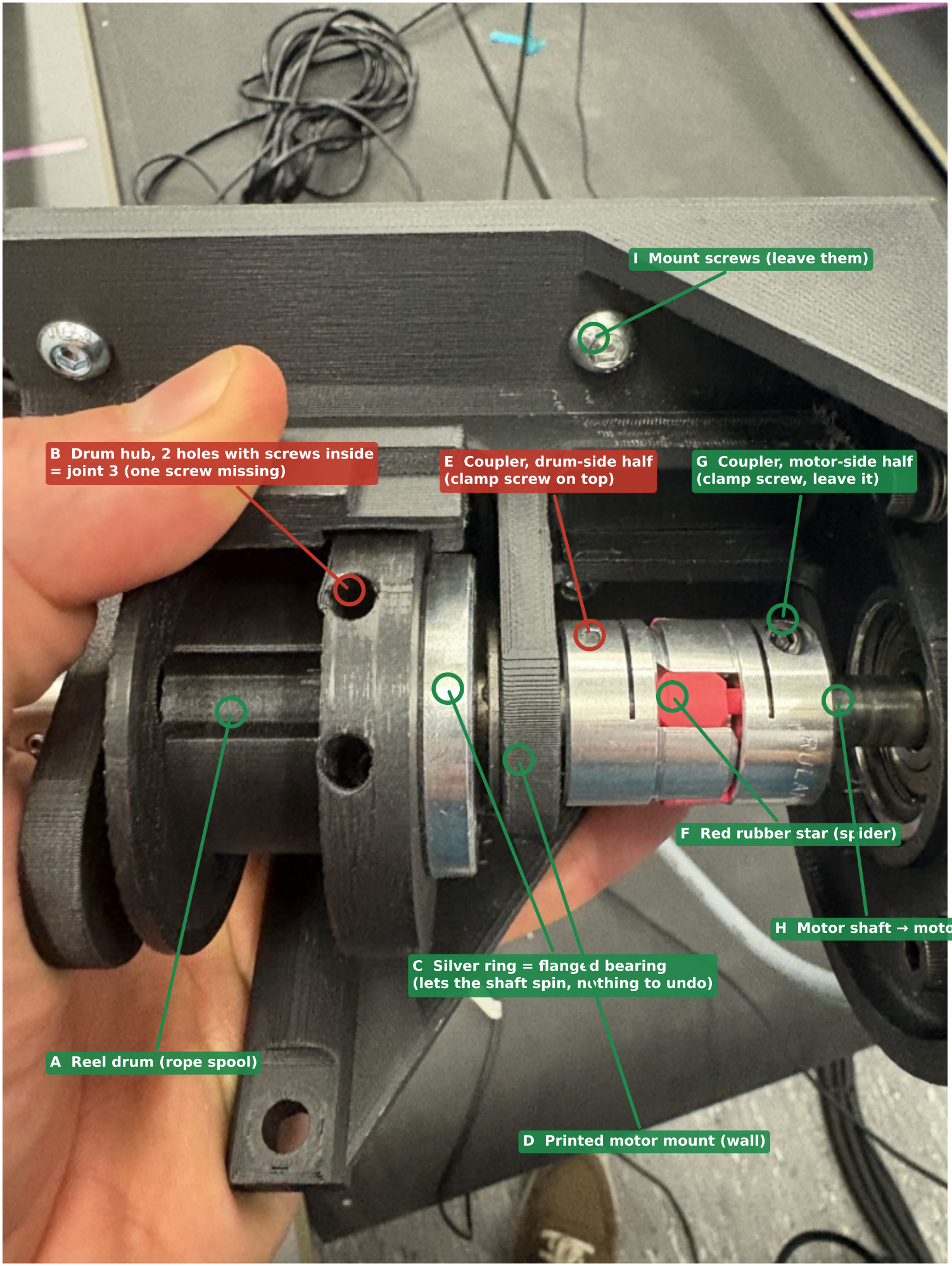
Your motor unit (tower B), part by part

Red = you work on it. Green = leave it alone.



Take the drum off

Why: one screw in the drum hub (B) is missing; the drum slipped on the shaft.

First try without taking it apart: a new screw through the empty hole (see last page).

Only take the drum off if that does not work.

0 Prepare

- UI: Release (treadmill stopped). Stop the server. This tower's 48 V and 24 V OFF.
- Take 4 photos (front, back, both sides). Put a cup or a magnetic tray next to you.
- Marker: one line across drum hub (B), shaft and coupler (E), so it goes back the same way.

1 Free the rope

- Unhook the rope from the load cell / dummy. Unwind it from the drum.
- The knot sits inside the drum core: push it out through the tie hole, or leave the rope on and take drum + rope off together.

2 Loosen the drum hub screws (B)

- Turn the drum until a hole faces you. Hex key: inch, 3/32" or 7/64" (whichever fits with no wobble). Turn counter-clockwise until the screw is loose. Same for the other hole.
- You do not have to take them out. The drum should now turn on the shaft by hand.

3 Loosen the coupler's drum-side clamp screw (E)

- The small screw on top of the silver coupler half next to the black mount wall.
  Size: try metric 2 / 2.5 mm and inch 3/32" / 7/64". Two turns are enough.
- Do NOT touch the motor-side half (G): it stays on the motor shaft.

4 Pull the shaft out

- Pull the drum (with the shaft inside) away from the mount, straight, twisting a little.
  The shaft slides out of the coupler (E) and the bearing (C).
- Or, if the shaft stays: slide the drum off the shaft end.
- **It does not move with firm hand force? STOP. Something on the far side of the drum holds it (the big black disc, left of A). Send a photo of that side before going on.**
- **Never hammer: printed parts crack.**

5 Lay the parts out in order

- Drum, any washers or spacers (note where they were), shaft. The red star (F) may fall out of the coupler: put it back into the motor-side half (G).

Fix and put it back

6 While it is apart

- Look at the shaft where the hub screws press: marks or a dent = where they slipped.
- File a FLAT there (10 mm long, ~0.5 mm deep; shaft in a vise). Clean away all filings.
- Shake the drum: the missing screw may be inside the hub. Get it out.
- New screws: take the remaining hub screw to a hardware store as a sample (likely #8-32 or #10-32 set screw, cup point). Buy 3.

7 Put it back (reverse order)

- Shaft through the bearing (C) into the coupler half (E), until it sits as deep as before (marker line). Keep ~1 mm gap between the red star and the claws: do not squeeze them.
- Tighten the coupler clamp screw (E) firmly.
- Drum onto the shaft as before (marker line). Turn it: it must not rub the mount or cover.
- Hub screws (B): one drop of blue threadlocker each, screw onto the FLAT, tighten firmly.

8 Test (after 10 min for the threadlocker)

- New marker line across hub, shaft and coupler.
- Rope back on: at least 3 turns stay on the drum when tight.
- 24 V + 48 V on, testing page: 30 N, 50 N, 100 N, 1000 ms each.
- Line still straight and the pulls get stronger = fixed. Then check towers A, C, D too.

Without taking it apart (try this first)

- Take the remaining hub screw out, buy matching ones, drop a new screw into the empty hole on the end of the hex key, threadlocker on both, tighten both.

Labels on page 1 are read from one photo of tower B.
If your tower looks different, send a photo and the guide is updated.